Galatea: Supplementary Material

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Equation, proposition and section numbers without the prefix S refer to the main paper.

APPENDIX: PROOFS

Proof of Proposition 1: The objective of (4) is strictly concave: the entropy is strictly concave, and the penalty $-\tau\,\mathrm{KL}(\mathbf{P}^\top\mathbf{1}\|\mathbf{b})$ is concave because KL is convex in its first argument and $\mathbf{P} \mapsto \mathbf{P}^\top\mathbf{1}$ is linear. The feasible set $\{\mathbf{P} \geq 0,\ \mathbf{P}\mathbf{1} = \mathbf{a}\}$ is a product of scaled simplices, hence non-empty, convex and compact, and the objective is continuous on it (with $0\log 0 = 0$), so a maximiser exists; by strict concavity it is unique. Because $\partial(-P\log P)/\partial P \to +\infty$ as $P \to 0$, the maximiser has $P_{ij} > 0$, so only the equality constraints are active. With multipliers f_i for the row constraints (Lagrangian term $\sum_i f_i(\sum_j P_{ij} - a_i)$) and $\mathbf{q} = \mathbf{P}^\top\mathbf{1}$, stationarity gives $\bar{S}_{ij} - \varepsilon\log P_{ij} - \tau\log(q_j/b_j) + f_i = 0$, that is, $P_{ij} = \exp\big((\bar{S}_{ij} + f_i + g_j)/\varepsilon\big)$ with $g_j := \tau\log(b_j/q_j)$. Summing over i gives $q_j = e^{g_j/\varepsilon}\sum_i e^{(\bar{S}_{ij}+f_i)/\varepsilon}$. Substituting into the definition of g_j gives $g_j(1 + \tau/\varepsilon) = \tau[\log b_j - \mathrm{LSE}_i((\bar{S}_{ij} + f_i)/\varepsilon)]$, which is (6) with $\lambda = \tau/(\tau+\varepsilon)$. The row constraint $\sum_j P_{ij} = a_i$ is (5). (i) Let Ψ be the composed map of Proposition 2, which is λ-Lipschitz. At the fixed point, $\|\mathbf{g}^\star\|_\infty \leq \|\Psi(\mathbf{g}^\star) - \Psi(\mathbf{0})\|_\infty + \|\Psi(\mathbf{0})\|_\infty \leq \lambda\|\mathbf{g}^\star\|_\infty + \|\Psi(\mathbf{0})\|_\infty$, and $\|\Psi(\mathbf{0})\|_\infty = O(\lambda)$ by (6). Hence $\|\mathbf{g}^\star\|_\infty \leq \|\Psi(\mathbf{0})\|_\infty/(1-\lambda) = O(\lambda)$, so $\mathbf{g}^\star \to \mathbf{0}$ as $\lambda \to 0$. Substituting $\mathbf{g} = \mathbf{0}$ into (5) gives $P_{ij} = a_i e^{\bar{S}_{ij}/\varepsilon}/\sum_k e^{\bar{S}_{ik}/\varepsilon}$. At $\tau = 0$ this holds exactly, because the problem decouples over rows into independent entropy-regularised linear programs on scaled simplices. (ii) As $\tau \to \infty$, $\tau\,\mathrm{KL}(\cdot\|\mathbf{b})$ Γ-converges to the indicator of $\{\mathbf{P}^\top\mathbf{1} = \mathbf{b}\}$. Since the feasible set is compact and the maximisers unique, $\mathbf{P}^\tau \to \mathbf{P}^{\mathrm{bal}}$ [1]. Setting $\lambda = 1$ in (5)–(6) gives the balanced Sinkhorn equations. □

Proof of Proposition 2: For any vector $\mathbf{x}$ and perturbation $\boldsymbol{\delta}$, $|\mathrm{LSE}(\mathbf{x} + \boldsymbol{\delta}) - \mathrm{LSE}(\mathbf{x})| \leq \|\boldsymbol{\delta}\|_\infty$, because $\nabla\mathrm{LSE}$ is a probability vector. Hence F is 1-Lipschitz in $\|\cdot\|_\infty$, G is λ-Lipschitz, and $\Psi = G \circ F$ is λ-Lipschitz. For $\lambda < 1$, Ψ is a contraction on $(\mathbb{R}^{m+1}, \|\cdot\|_\infty)$, so by Banach's theorem it has a unique fixed point $\mathbf{g}^\star$ (the potential of Proposition 1), and $\mathbf{g}^\star = G(\mathbf{f}^\star)$ with $\mathbf{f}^\star = F(\mathbf{g}^\star)$. Algorithm 1 computes $\mathbf{g}^{(l)} = G(\mathbf{f}^{(l-1)})$ and $\mathbf{f}^{(l)} = F(\mathbf{g}^{(l)})$ for $l = 1, \ldots, L$. Therefore $\|\mathbf{g}^{(l)} - \mathbf{g}^\star\|_\infty \leq \lambda\|\mathbf{f}^{(l-1)} - \mathbf{f}^\star\|_\infty$ and $\|\mathbf{f}^{(l)} - \mathbf{f}^\star\|_\infty \leq \|\mathbf{g}^{(l)} - \mathbf{g}^\star\|_\infty$; chaining these inequalities gives the bound. The rate does not depend on ε, n or m, because the Lipschitz constants do not. □

Proof of Proposition 3: After (5) with $a_i = 1$, $e^{f_i/\varepsilon} = 1/\sum_{j\leq m+1} e^{(\bar{S}_{ij}+g_j)/\varepsilon}$, so $P_{i,m+1} = e^{(z+g_{m+1})/\varepsilon}\big/\big(e^{(z+g_{m+1})/\varepsilon} + \sum_{j\leq m} e^{(s_{ij}+g_j)/\varepsilon}\big)$, which equals (7). The argument of the sigmoid is strictly decreasing in each s_{ij}, because $\partial\,\mathrm{LSE}/\partial x_j > 0$, and increasing in z. □

TABLE SI
BACKBONE LEARNING RATE FOR JOINT FINE-TUNING, SELECTED ON THE VALIDATION SCENE (GALATEA HEAD, SEED 0, R@1 %).

Backbone lr	$1\cdot10^{-5}$	$3\cdot10^{-5}$	$1\cdot10^{-4}$	$3\cdot10^{-4}$	$1\cdot10^{-3}$
Val. R@1	68.4	70.4	76.6	68.9	18.4

Proof of Proposition 4: By (9), $\mathbf{V}_{j,:} = \sum_i P_{ij}\mathbf{h}_i$. Replacing P_{ij} by $c_j P_{ij}$ multiplies row j of $\mathbf{V}$ by $c_j > 0$, which intra-normalisation removes. The scene token does not depend on $\mathbf{P}$, so Φ is unchanged. For $P_{ij} = e^{(\bar{S}_{ij}+f_i+g_j)/\varepsilon}$, take $c_j = e^{-g_j/\varepsilon}$ to obtain (8), with $w_i = e^{f_i/\varepsilon}$ given by (5). For $\lambda = 0$, $\mathbf{g} = \mathbf{0}$ and $a_i = 1$. □

Proof of Proposition 6: $s(q,r) > s(q,r')$ is equivalent to $\beta\Delta > \delta$, which is (i). Since $0 \leq \phi \leq \phi_{\max}$, $\beta\Delta \leq \beta\phi_{\max}$, which gives (ii). Since ϕ is increasing, $s_\ell(q,r) \leq s_\ell(q,r')$ implies $\Delta \leq 0 \leq \delta$, which gives (iii). □

Proof of Proposition 5: For a similarity transform $\mathbf{u} \mapsto c\mathbf{R}\mathbf{u} + \mathbf{t}$, all pairwise distances scale by c and so does their mean μ, which leaves $\mathbf{C}^q$ unchanged. The appearance cost $\mathbf{M}$ does not depend on positions, and neither does the dustbin cost. This proves the claim for (10). The solver uses the positions only through $\mathbf{C}^q$ and $\mathbf{C}^c$ in (11), and its initial plan depends on $\mathbf{M}$ alone, so every iterate, the mutual-arg-max correspondences and their number are unchanged as well. A permutation of keypoints permutes rows and columns of $\mathbf{M}$, $\mathbf{C}$, $\mathbf{G}^{(\nu)}$ and $\mathbf{T}$ consistently; Sinkhorn scaling commutes with such permutations, so every iterate and the correspondences are permuted accordingly and their number is unchanged.□

REFERENCES

[1] L. Chizat, G. Peyré, B. Schmitzer, and F.-X. Vialard, "Scaling algorithms for unbalanced optimal transport problems," *Math. Comput.*, vol. 87, no. 314, pp. 2563–2609, 2018.

SI. ADDITIONAL EXPERIMENTS

Backbone learning rate: For joint fine-tuning (Section VI-D) the backbone learning rate was selected on the validation scene only, with the Galatea head and seed 0 (Table SI). The head learning rate, schedule, sampler and loss are those of the frozen-backbone runs; training lasts 1000 steps. Test results were not consulted. The largest rate diverges.

Cross-hardware reproduction: All frozen-backbone runs of the main paper were executed on a 2-core CPU. We re-ran them with identical code on NVIDIA Tesla P40 GPUs, after re-extracting the frozen features on the GPU (the zero-shot baselines reproduce exactly: 26.8, 28.1 and 35.8% R@1), and added two seeds for the four main heads (Table SII). All means agree within one standard deviation, and every conclusion

TABLE SII
CROSS-HARDWARE REPRODUCTION OF THE FROZEN-BACKBONE HEADS (TEST R@1, %, MEAN±STD). CPU: THE RUNS REPORTED IN THE PAPER (3 SEEDS). GPU: THE SAME CODE AND PROTOCOL RE-RUN ON NVIDIA TESLA P40 GPUS (5 SEEDS).

Head	CPU (3 seeds)	GPU (5 seeds)	GPU hard
GeM	31.5±0.4	32.3±1.1	22.1
NetVLAD	58.2±0.7	58.1±0.6	46.6
SALAD (λ=1)	59.4±0.7	60.0±0.6	48.8
Galatea, learned λ	59.8±0.8	59.7±1.0	47.8
$\lambda = 0$	59.9±0.8	60.1±0.9	48.1

of the paper holds. The single-seed ablations also reproduce: single-illumination training gives 47.8% (CPU: 47.3%).

SII. ROBUSTNESS TO CAMERA TILT AND VIEWPOINT

The benchmark queries differ from the database by small viewpoint changes only: at most 3 m of offset, heading noise $\mathcal{N}(0, 10^\circ)$, camera heights of 1.35–1.7 m, and no roll or pitch. Here we change one factor at a time, starting from the exact benchmark query poses, and re-render all 3150 test queries under all seven Sun conditions. The database stays the original upright one (Fig. S1, Table SIII).

Protocol: LunarSim casts rays column by column, which is exact only for an upright camera. A camera rotated about its centre sees the same rays re-projected, so its image is an upright image with the same focal length warped by the homography $\mathbf{KRK}^{-1}$, independently of scene depth. We render the upright source at $3\times$ resolution, keeping the micro-relief level of detail of the native resolution, and point-sample it through the homography. Without tilt this reproduces the benchmark renderer exactly; a 0.01° roll differs from the native image by 0.15 grey levels on average.

- *Roll* is a rotation about the camera's central viewing direction, so an upside-down camera sees the same scene upside down.
- *Gravity correction* simulates levelling the image with an IMU: the upright view, black outside the part of the scene the rolled camera sees.
- *Offset* fixes the distance to the database place exactly, in the benchmark's random direction.
- *Heading* adds a change of random sign to the benchmark heading. At $+45^\circ$, 70 queries have no true match within 10 m and 60° and are excluded.

The frozen heads are the GPU runs of Table SII. The fine-tuned runs were retrained with the protocol of Section VI-D so that their backbones could be stored; they reach 71.3 and 77.0% R@1 upright (Table II: 71.3 and 76.4). On the re-rendered upright queries, frozen Galatea reaches 59.8%.

Roll: Roll is the most damaging factor. A 1° roll already costs frozen Galatea 4.0 points (paired 95% CI $[2.4, 5.7]$), 5° leaves 41.1% and 15° 19.5%. From 45° on, no model exceeds 10%; upside down, frozen Galatea reaches 7.8% and fine-tuned NetVLAD 5.4%. Fine-tuning does not remove the sensitivity: fine-tuned NetVLAD falls from 77.2 to 61.3% at 5°. No training or database image is rolled, and DINOv2 tokens are not rotation-invariant. Levelling the image with gravity recovers most of the loss. Upside down, it brings frozen Galatea back to 46.9% and fine-tuned NetVLAD to 72.4%. The remaining cost, 11–26 points for frozen Galatea and 5–18 for the fine-tuned models, comes from the black regions outside the rolled frame.

Pitch: Pitching by $\pm 5^\circ$ costs frozen Galatea up to 6.6 points. Looking 30° down, the frame shows only ground within a few metres and no horizon; no model exceeds 4.0%. Part of this failure may be due to the simulator, whose near-field ground is limited by the 0.3 m DEM. Looking 30° up, the frame is mostly sky. Galatea then beats NetVLAD both with a frozen backbone (32.1 vs. 20.9%) and with a fine-tuned one (43.2 vs. 33.5%). This is consistent with the dust-bin acting as a gate on sky tokens (Proposition 3), although the two heads also differ in their aggregation.

Viewpoint: Moving 8 m away from the database place costs frozen Galatea 21 points relative to 0 m. Adding 15° or 30° of heading costs 17 and 42 points. Each place is in the database with a single heading, and most of the frame shows nearby ground, which changes completely when the rover turns. Camera heights of 1.0 and 2.5 m (database: 1.5 m) cost 9–12 points, and 0.5 and 4.0 m 26–31.

Fusion: Under small tilts and offsets, fusion still adds 4–12 points. Once the correct place leaves the ten candidates it cannot help. Under roll, MNN+MAGSAC++ fusion degrades less than FGW (35.5 vs. 23.4% at 15°): FGW's structural term is rotation-invariant (Proposition 5), but its appearance term is not.

Implications: A rover should level its images with the IMU before retrieval. Its database should be recorded at the operating camera height and with several headings per place. Offsets of a few metres cost little (5 points at 4 m).

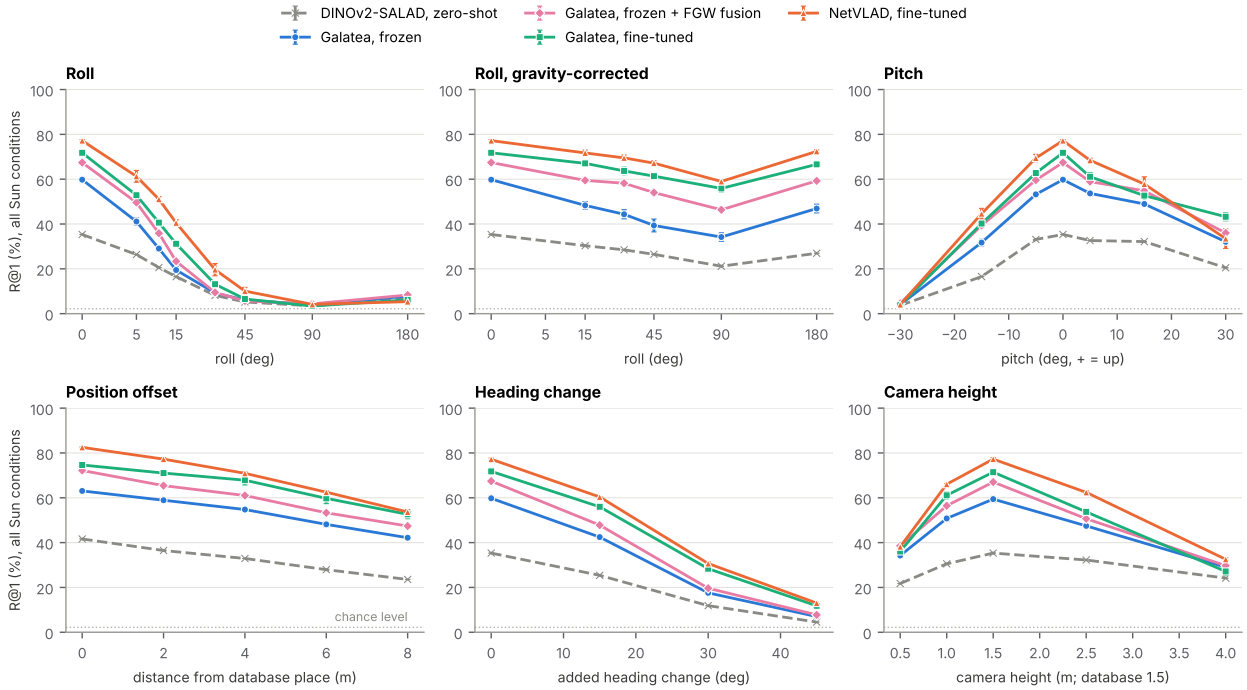


Fig. S1. Recall@1 over all seven Sun conditions when one factor of the query camera is changed (3 seeds, error bars: std). Roll axes are square-root scaled; dotted line: chance (2.2%).

TABLE SIII

ROBUSTNESS TO CAMERA TILT AND VIEWPOINT: R@1 (%) OVER ALL SEVEN SUN CONDITIONS ON THE TEST SET (3150 QUERIES PER ROW; CHANCE 2.2%). EACH ROW RE-RENDERS THE BENCHMARK QUERIES AT THE BENCHMARK POSES WITH ONE FACTOR CHANGED; THE DATABASE IS THE ORIGINAL UPRIGHT ONE. MEAN±STD OVER 3 SEEDS; FUSION ON THE FROZEN GALATEA HEAD, SEED 0, WITH THE VALIDATION-TUNED (β, κ) OF THE MAIN PAPER.

	Zero-shot	Frozen backbone		Fine-tuned backbone		Frozen Galatea + fusion	
Variant	SALAD	Galatea	NetVLAD	Galatea	NetVLAD	FGW	MNN+MAGSAC++
Upright (baseline)	35.3	59.8±1.1	58.4±0.5	71.8±1.2	77.2±0.5	67.4	68.2
Roll 1°	33.0	56.1±0.8	54.3±0.3	66.6±1.3	73.6±0.7	61.7	64.6
Roll 5°	26.4	41.1±1.5	43.6±0.8	52.9±0.9	61.3±2.6	49.6	54.9
Roll 15°	16.5	19.5±1.6	23.1±0.9	31.2±0.8	40.5±1.4	23.4	35.5
Roll 45°	5.2	6.1±0.4	5.6±0.7	6.5±0.6	10.1±1.7	5.7	6.8
Roll 180°	6.9	7.8±0.3	5.3±0.3	6.1±0.5	5.4±1.1	8.3	8.2
Roll 45°, gravity-corr.	26.5	39.4±2.9	39.3±1.6	61.4±1.2	67.2±1.1	54.0	57.3
Roll 180°, gravity-corr.	26.9	46.9±1.9	46.5±1.2	66.6±0.5	72.4±0.7	59.3	63.0
Pitch −30°	3.7	4.0±0.3	4.0±0.3	3.8±0.3	4.0±0.5	4.6	4.7
Pitch −15°	16.5	31.7±1.5	29.3±1.5	40.2±0.4	44.6±2.4	39.4	43.6
Pitch +15°	32.2	49.0±0.9	48.2±0.6	52.7±3.3	57.8±3.2	54.8	56.9
Pitch +30°	20.5	32.1±0.3	20.9±1.4	43.2±1.8	33.5±4.4	36.0	42.0
Offset 0 m	41.6	63.1±0.6	66.7±0.4	74.6±1.3	82.5±0.3	72.2	72.8
Offset 4 m	32.9	54.8±0.8	51.4±0.4	67.8±2.1	70.9±0.6	61.0	62.5
Offset 8 m	23.6	42.2±1.0	37.8±0.4	52.6±2.0	53.6±0.9	47.4	47.6
Heading +15°	25.4	42.5±1.1	40.0±0.1	56.0±1.1	60.4±0.2	47.8	49.3
Heading +30°	11.9	17.6±0.6	16.8±0.4	28.4±0.6	30.7±0.7	19.6	20.4
Height 0.5 m	21.8	34.2±0.9	33.0±0.3	36.0±1.7	38.2±0.5	38.5	38.2
Height 1.0 m	30.5	50.8±1.3	50.5±1.2	61.1±1.6	66.0±0.4	56.4	56.6
Height 2.5 m	32.2	47.5±0.5	48.4±0.9	53.7±1.4	62.4±0.7	50.7	52.4
Height 4.0 m	24.2	29.0±1.4	30.9±0.6	27.1±1.8	32.5±0.7	29.7	29.1